

Dehydration: Physiology, Assessment, and Performance Effects

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ABSTRACT

This article provides a comprehensive review of dehydration assessment and presents a unique evaluation of the dehydration and performance literature. The importance of osmolality and volume are emphasized when discussing the physiology, assessment, and performance effects of dehydration. The underappreciated physiologic distinction between a loss of hypo-osmotic body water (intracellular dehydration) and an iso-osmotic loss of body water (extracellular dehydration) is presented and argued as the single most essential aspect of dehydration assessment. The importance of diagnostic and biological variation analyses to dehydration assessment methods is reviewed and their use in gauging the true potential of any dehydration assessment method highlighted. The necessity for establishing proper baselines is discussed, as is the magnitude of dehydration required to elicit reliable and detectable osmotic or volume-mediated compensatory physiologic responses. The discussion of physiologic responses further helps inform and explain our analysis of the literature suggesting a $\geq 2\%$ dehydration threshold for impaired endurance exercise performance mediated by volume loss. In contrast, no clear threshold or plausible mechanism(s) support the marginal, but potentially important, impairment in strength, and power observed with dehydration. Similarly, the potential for dehydration to impair cognition appears small and related primarily to distraction or discomfort. The impact of dehydration on any particular sport skill or task is therefore likely dependent upon the makeup of the task itself (e.g., endurance, strength, cognitive, and motor skill). © 2014 American Physiological Society. *Compr Physiol* 4:257-285, 2014.

Introduction

Dehydration is a word commonly understood to describe a deficit in total body water (TBW). Body fluids also contain dissolved substances, the transient loss of which can alter the redistribution of water across cell membranes. When the concentration of lost body fluids is hypo-osmotic with respect to plasma, an osmotic gradient is created and the partitioning of water loss is shared between the intracellular fluid space (ICF) and extracellular fluid space (ECF) (69, 76, 175, 215, 217, 232, 258, 268, 284, 382, 383). The relatively large loss of ICF and the hyperosmotic character of the ECF space are often referred to as “intracellular dehydration,” or sometimes hyperosmotic-hypovolemia, hypohydration, or dehydration with minimal salt loss. When the concentration of lost body fluids is iso-osmotic with respect to plasma, a relatively larger contraction of the ECF space occurs (extracellular dehydration), the osmotic gradient to pull fluids from the larger intracellular reservoir is absent, and the character of the ECF is unremarkable (23, 74, 76, 258). “Extracellular dehydration” is also sometimes referred to as iso-osmotic-hypovolemia, volume depletion, or dehydration with salt-loss. The important physiologic distinction between the two major “types” of dehydration, though not a new concept (212, 214, 258), is a driving distinction for treatment (172, 234) and arguably the most essential and underappreciated aspect of dehydration assessment (58).

The physiologic response to intracellular dehydration arises from the primary stimulation of osmoreceptors and secondary stimulation of baroreceptors (91, 293). Endocrine control of water balance includes hormone-mediated compensatory renal water conservation and water acquisition (thirst) (12). The response to extracellular dehydration is nonosmolar (318, 319) and includes an acute sympathetically mediated vasoconstriction and slower neuro-humoral defense of blood volume (BV) (317, 378). The osmolality (mmol/kg) and volume thresholds for these responses depend importantly on the “type” of dehydration that has occurred and the magnitude of dehydration required to elicit compensation (58). The type and magnitude of dehydration also inform the physiology which underpins the predominantly negative effects of dehydration on some types of human performance.

Previous Handbook of Physiology articles in *Comprehensive Physiology* have provided detailed reviews of body fluid balance in austere environments (182, 230), as well as the complete endocrine regulation of body water (12) and sodium balance (117). Although the subject of dehydration

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assessment and the effects of dehydration on exercise performance were both treated briefly within the broader context of a recent *Comprehensive Physiology* review of integrated and adaptive physiology in the heat (311), no review in this historical series provides an extensive treatment of these subjects. This article will, therefore, (i) review the physiology of dehydration, (ii), provide historic and novel insights into the science of dehydration assessment, and (iii) offer a unique analysis of the evidence for and against the effects of dehydration on human performance to include plausible mechanisms.

Physiology of Dehydration

Historical backdrop

Systematic inquiry into the physiology of dehydration began in the 17th century with the quantitative characterization of water loss from the human body (28). At the turn of the 20th century, McGee (244) published the first qualitative account of graded signs and symptoms associated with dehydration in the desert southwest. It was not until after World War I, however, that the complexity of ECF was recognized and the discipline of body fluid physiology was born under the tutelage of James Gamble (174). By 1923, the concentrating effects of dehydration on the blood and the added strain that it imposed on the circulation were well documented (237), as was the maximal water excretion potential of the kidneys (5). In the 1930s, methods of measuring body water losses from the stoichiometric sum of direct laboratory observations and indirect calculations (265) developed in parallel with Daniel Darrow's characterization of osmotic equilibrium between ICF and ECF spaces (76). Osmometric formulae were also published at this time to help conceptualize water exchanges within the body and between body and environment (215). The important physiologic distinction and differences in treatment between intracellular and extracellular dehydration soon followed (258).

World War II brought renewed interest in optimal ration development and study of the minimal water requirements for survival. The minimal renal water requirements of man were gleaned from life raft (castaway) studies (123–125) to complement earlier work on maximal renal water excretion (5). Adolph (7) added a quantitative element to McGee's qualitative dehydration symptom scale in conjunction with the most comprehensive series of studies describing dehydration's effects on human physiology before 1950 (or since). In the same decade, isotopes of hydrogen or oxygen were being considered for the first time to accurately measure TBW in man (256) and Verney described the osmoreceptor and attributed the constancy of osmotic pressure of body fluids to arginine vasopressin (AVP) (361). By the late 1950s, Edelman (94) provided an important equation for use by physiologists and clinicians that afforded quantitative assessment of water and electrolyte mass balance. The roles of aldosterone and baroreceptors in antidiuresis and antinatriuresis became well recognized, though no consensus had yet been reached

concerning the location of the baroreceptors (23, 334). Just before 1960, estimations for the size and composition of body water compartments were reviewed (93) and still reflect remarkably well the values with which we are today familiar. Improvements to the measurement of fluid regulatory hormones in the 1970s (294) made possible a better understanding of the relative roles of osmotic and volume (baroreceptor) contributions to the regulation of water metabolism (91, 295, 318, 319). These key discoveries and advances of the early 20th century are not all-inclusive, but they represent important highlights for understanding the fundamental physiology, assessment, and performance effects of dehydration.

Body water and body mass thresholds

The percentage of body mass as water is commonly taken to be 60%, but it is acknowledged that this value varies in accordance with body composition. TBW can therefore be better estimated as a constant fraction (~ 0.737) of fat-free mass (368). Isotope dilution is an often referenced gold standard for TBW determination since the accuracy of the method closely approximates values measured by desiccation and its analytical measurement variation (CVa) is as small as 1% to 2% (~ 0.4 – 0.8 L) (316). A single isotope dilution measurement cannot be used to assess dehydration due to the aforementioned variation in body composition among people. The smallest acute loss of TBW detectable by dilution is 0.6 to 1.1 L ($CVa \times \sqrt{2}$) when measuring a change. In contrast, the use of dilution in conjunction with a standard medical grade scale can reduce the detection threshold for acute TBW losses by an order of magnitude (≤ 0.05 kg). In fact, when knowledge of a "true" dehydrated status is considered crucial for testing diagnostic procedures (338), an acute loss of body mass may be considered synonymous with an acute loss of TBW (1 kg = 1 L) so long as water loss (sweat, urine, and ultrafiltrate) (21, 71, 300) or water re-gain from an unknown premonitory state (i.v. or fluid intake) (285, 343) is observed. The percentage of body mass that is dehydrated may then be calculated as: %dehydration = $(\Delta \text{ body mass} / \text{initial body mass}) \times 100$. It is important to note that studies in which longer observations periods are required, or those in which food and fluid is consumed, require additional considerations for measuring all avenues of total water and solid exchange (63, 265). Similarly, this technique is reliant on individuals remaining in energy balance throughout any given observation period. Whether or not the baseline body mass (or TBW) from which change is calculated reflects a typical (normally hydrated) condition is an important separate question that is sometimes difficult to answer (212, 334). However, this question is of critical importance when attempting to determine the impact of a specific level of dehydration on physiology or performance outcomes.

One way to establish an adequate baseline is to make serial measurements of body mass from day-to-day. Adolph and Dill (4) monitored daily body mass changes while also making measures of water balance in seven men during the

Harvard Desert Expedition to Nevada in 1937 (hot weather). Similar observations were made on five of the same men while in Massachusetts (temperate weather) for comparison. The day-to-day within-person coefficient of variation (CVw) for body mass was 0.25% and 0.48% in temperate and hot weather environments, respectively. By extension, TBW must have deviated day-to-day by no more than ~0.2 to 0.4 L/day (4). The study of Adolph and Dill, part laboratory and part field, was necessarily contrived and may therefore have underestimated the true day-to-day variation in TBW. However, studies following similar (54) and less-controlled procedures (50), as well as those making daily body mass measures in free-living people (377) all report CVw values for body mass between 0.5% and 1%. Day-to-day body mass variation $\leq 1\%$ therefore provides one means of establishing a baseline from which observed losses (change) can be used to reliably quantify dehydration (1 kg = 1 L).

Day-to-day variation in body mass can also be used to estimate the probability that unobserved losses in body mass are attributable to dehydration. For example, a day-to-day change in body mass $>2\%$ provides 95% probabilistic certainty that dehydration has occurred, assuming a one tailed test of significance in the direction of biological plausibility (i.e., $\sqrt{2} \times 1.65$ when CVw = 1%) (54). A postmorbidity body weight that is 2% less than documented “well weights” might similarly be used in the absence of a valid premorbidity measurement (343). Therefore, day-to-day fluctuations of $\leq 1\%$ body mass represent normal variation. Unlike the precision made possible by the observation of acute losses, unobserved losses from day-to-day must exceed a threshold of 2% (~1.0–1.5 L) to be considered truly atypical. Daily fluctuations in body mass between 1% and 2% cannot be reliably associated with perturbations in TBW and should therefore be considered typical or well-hydrated (58, 212). For reasons to be discussed later, dehydration by $\geq 2\%$ body mass is also the threshold beyond which some types of human performance become measurably and reliably impaired (212, 309). Dehydration assessment by means other than body mass depends critically upon physiologic compensation thresholds for renal water conservation, water acquisition (thirst), and integrated adjustments to vascular control (12, 317, 378), all of which depends on the type and magnitude of dehydration incurred.

Intracellular dehydration thresholds

Intracellular dehydration occurs when the concentration of lost body fluids is hypo-osmotic with respect to plasma. The volume of hypo-osmotic sweat losses (<100 mmol/kg) (69, 379) that occur during exercise commonly range from 0.50 to 2.0 L/h and are proportional to body heat gained by metabolism and the environment (345). However, even in the absence of active sweat glands (thermoregulatory sweating), the loss of pure water by transcutaneous evaporation from the skin can range from 0.7 to 3.0 L/day, depending on the environment (37, 209, 264). When coupled with obligatory renal water losses, fluid restriction due to water unavailability

or all-cause adipsia will lead to intracellular dehydration, or the loss of body water with minimal losses of solute. Osmotic diarrhea is another cause of hypo-osmotic body water losses. Whereas an infectious agent produces secretory diarrhea that carries away both water and electrolyte (169), osmotic diarrhea occurs when water is pulled into the intestine by osmotic forces produced by any variety of poorly absorbed substances (158), such as lactose, gluten, laxatives, or sugar alcohols.

The conditions produced by dehydration must be contrasted from ordinary daily body water flux (noncompensatory). Osmotic constancy is ordinarily maintained by the secretion of AVP, which directly influences renal water excretion and conservation in response to ordinary body fluid shifts and *ad libitum* food and fluid intakes (281, 322, 330). Osmotic homeostasis corresponds with basal electrical activity of osmoreceptor neurons believed to encode an osmotic set-point (35) that ordinarily keeps plasma osmolality (Posm) from deviating more than 1–2% (12, 35, 294). Indeed, the CVw for Posm from day-to-day varies from 0.8% to 1.4% (2–4 mmol/kg), excluding CVa (~0.5% or 1 mmol/kg) (54, 116, 190). However, as hypo-osmotic body water losses increase the osmotic pressure of the ECF, the requirement for osmotic equilibrium between ICF and ECF spaces causes transmembrane flow of water from the larger ICF space (i.e., intracellular dehydration) (69, 76, 173, 215, 217, 232, 258, 268, 284, 382, 383). Water movement and osmotic equilibrium occur quickly (62, 167). Slower two way fluid flow within the ECF space itself, between interstitial (ISF) and plasma volume (PV) spaces, occurs due to nonosmotic shifts since these compartments share the same osmotic pressures (62, 164, 173, 268). Cellular shrinkage (dehydration) is sensed in the brain by osmoreceptors (361), which Bourque (34, 35) describes functionally as specialized neurons capable of sensing osmolality and encoding osmosensory information as electrical signals. An increase in neuronal firing frequency stimulates hypothalamic release of AVP, which concentrates urine by stimulating renal water reabsorption (12, 35, 361). Osmoreceptor-mediated thirst is independent of AVP and stimulated by direct neural signaling (35, 80, 246). There is also evidence that angiotensin II mediates osmotically driven thirst (12, 35, 246).

A ~2% increase in Posm (~6 mmol/kg) is commonly referenced as an osmotic threshold for compensatory renal water conservation and water acquisition (thirst) (12). The theoretical increase in Posm for any volume of hypo-osmotic TBW losses (76, 382, 383) is in good agreement with measured values. Posm increases by ≥ 5 mmol/kg in response to intracellular dehydration when TBW losses are $\geq 2\%$ of body mass (1.4 L at 70 kg) (22, 54, 55, 286, 328). The correspondence between these two thresholds for body mass and Posm changes affords $\geq 80\%$ probability that significant hypo-osmotic body water losses have occurred (55). Variation in sweat sodium losses among people may (268), or may not (313), add uncertainty to the magnitude of the osmotic response to a given water deficit, depending on the delicate balance between sweating rate and sweat sodium

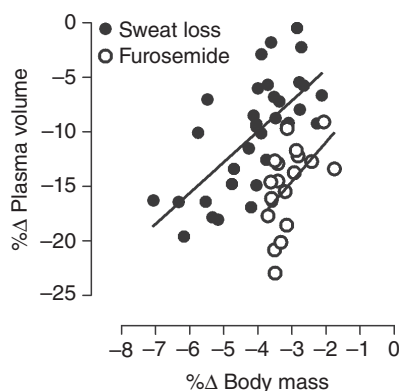


Figure 1 Linear regression of plasma volume change (y) and body mass change (x) following sweat loss or diuretic dehydration. Original data are shown for 18 volunteers (54 data points) dehydrated twice by $4.1 \pm 1.5\%$ using 3 to 5 h of exercise-heat exposure (sweat; $n = 36$) and once by $3.1 \pm 0.5\%$ using 40 and 80 mg diuretic (Furosemide; $n = 18$). Plasma volume contraction with Furosemide ($y_F = -3.8 + 3.6x$) is double that for sweat ($y_S = 1.35 + 2.8x$) at any given level of dehydration. Adapted, with permission, from Cheuvront et al. (56).

concentrations. However, we can estimate (76, 382, 383) that a $\pm 50\%$ change in sweat osmolality should produce marginal fluctuations in Posm (± 2 mmol/kg). Similar considerations may be made of alterations in extracellular volume, or PV specifically (268). However, based on the regression equations in Figure 1 (56) the anticipated decrease in PV is only ~ 0.14 L at $\sim 2\%$ dehydration. This is approximately half the PV loss expected for the same TBW loss in extracellular dehydration and entirely consistent with rapid osmotic redistribution of water in intracellular, but not extracellular dehydration. Intracellular dehydration therefore results in a small ratio of PV to TBW losses ($\sim 1:10$), though the precise ratio may vary according to the magnitude (small vs. large losses) and mode (passive vs. active) of dehydration (184, 315). Although the sensitivity of osmoreceptors in regulating AVP release is enhanced by relatively small losses of volume, it is unlikely that volume-mediated thirst is triggered during intracellular dehydration given the larger required threshold loss of $\sim 10\%$ BV (12, 91, 246, 293), which is ~ 0.50 L. A summary of osmotic water regulation in response to intracellular dehydration is depicted in Figure 2 (12, 58).

Extracellular dehydration thresholds

When compared to intracellular dehydration, extracellular dehydration is characterized by an unremarkable ECF osmolality but larger ECF volume contraction (76, 234, 258). This occurs because a loss of body water that is iso-osmotic with plasma requires a significant accompanying loss of electrolytes. The most common example of extracellular dehydration occurs in secretory diarrhea when enteric pathogens stimulate the secretion of ions into the gut (169). The iso-osmotic character and larger contraction of ECF in infectious diarrheal disease is well-documented (48, 74, 75, 175, 173). It is also well-documented (176) that acute arterial hypoxemia

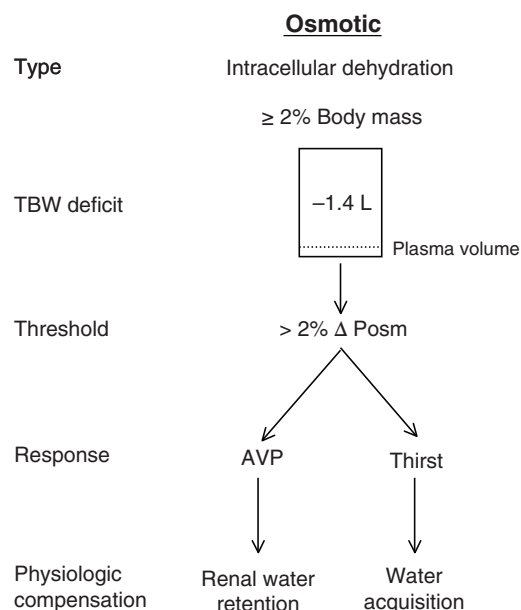


Figure 2 Osmotic regulation of body water balance in response to intracellular dehydration. Schematic includes the estimated magnitude of dehydration required to stimulate the osmotic-dependent response for compensatory water conservation and acquisition. TBW is total body water, Posm is plasma osmolality, AVP is arginine vasopressin. A change in TBW is equated with a change in body mass ($1 \text{ L} = 1 \text{ kg}$), whereby dehydration is then expressed as a percentage of body mass in accordance with: $(\Delta \text{ body mass} / \text{initial body mass}) \times 100$. The PV:TBW volume ratio is depicted to scale (1:10). Composite figure created, with permission, from content in Andreoli et al. (12) and Cheuvront et al. (58).

produces diuresis and natriuresis. The resulting loss of water and ions produces a large, iso-osmotic loss of ECF volume which serves to rapidly concentrate hemoglobin and improve arterial oxygen tension. Although hypoxia, rather than hypobaria, is the stimulus for the inhibition of renal tubular sodium reabsorption (176), humoral, and other factors are also likely involved (31, 156, 176, 182). Similarly, more than one factor (neural, humoral, and hemodynamic) contributes to the diuresis and natriuresis observed in response to cold exposure (219).

Extracellular dehydration can also be produced without infectious agents or extreme environments. For example, loop diuretics like Furosemide (Lasix) produce similar losses of water and ions by blocking the reabsorption of sodium in the ascending loop of Henle (298, 339). Although their intended use is for removing excess body water in association with edema and hypertension, acute doses of 40 to 80 mg can be given to normally hydrated people to produce iso-osmotic-hypovolemia of $\sim 3\%$ body mass with a ratio of PV:TBW loss (1:5) double what occurs for the same level of intracellular dehydration (Fig. 1) (56). Head-out water immersion can be used for similar purposes. The hydrostatic pressure created by water increases central BV and blood pressure, triggering a cascade of sympathetic and neurohumoral responses that produce both copious diuresis and natriuresis (102) that

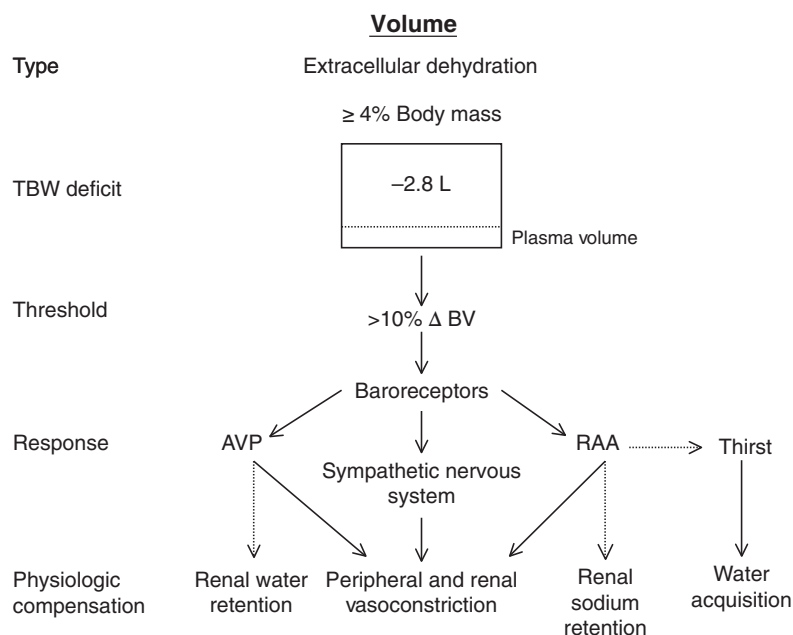


Figure 3 Volume regulation of body water balance in response to extracellular dehydration. Schematic includes the estimated magnitude of dehydration required to stimulate the volume-dependent response for compensatory water conservation and acquisition. Solid lines represent primary and dashed lines secondary responses. TBW is total body water, AVP is arginine vasopressin, RAA is rennin-angiotensin-aldosterone. A change in TBW is equated with a change in body mass (1 L = 1 kg), whereby dehydration is then expressed as a percentage of body mass in accordance with: $(\Delta \text{ body mass}/\text{initial body mass}) \times 100$. The PV:TBW volume ratio is depicted to scale (1:5). Composite figure created, with permission, from content in Schrier (317), Andreoli et al. (12), and Cheuvront et al. (58).

is independent of water temperature from 25 to 35°C (204). It is, therefore, apparent that there are safe and useful methods for modeling extracellular dehydration when studying the potential efficacy of any extracellular dehydration assessment method (53, 272, 320).

Nonosmotic stimuli are responsible for compensatory responses to extracellular dehydration. The response to extracellular dehydration involves an acute, sympathetically mediated vasoconstriction and slower, neuro-humoral defense of BV (317, 378). Figure 3 (12, 58, 317) depicts the role of baroreceptors and the sympathetic nervous system in producing peripheral and renal arterial vasoconstriction. This is accomplished directly by the actions of norepinephrine, activation of the renin-angiotensin-aldosterone system, and nonosmotic stimulus of AVP. Renal reabsorption of water and sodium occur more slowly, but when superimposed on renal vasoconstriction it acts synergistically to restore circulatory integrity (317). An appreciation for the multiple interactive inputs from baroreceptors, the sympathetic nervous system, and the dual actions of fluid regulatory hormones to a volume challenge (317, 378) informs possibilities for assessing extracellular dehydration. However, in contrast to intracellular dehydration, the iso-osmotic nature of extracellular dehydration negates reaching any osmotic thresholds for compensation. We can, however, estimate based on the regression equations provided in Figure 1 that extracellular dehydration

by ~4% body mass (2.8 L at 70 kg) would be required to exceed the BV threshold of 10%, or approximately 0.50 L.

Mixed dehydration

It is important to acknowledge that the presentation of dehydration may very often be mixed. For example, the loss of both body water and solute anticipated with acute altitude exposure can be mixed with larger losses of pure water as altitude exposure becomes chronic. Solute-free respiratory water losses are typically balanced by metabolic water production across a range of activities at sea level (63, 248, 265). However, the combination of reduced barometric and water vapor pressures at altitude, coupled with increased respiration, lead to comparatively higher pure respiratory water losses (181). Chronic altitude exposure may also simultaneously alter osmotic and volume-mediated control of TBW balance (31). The loss of hypo-osmotic body fluids can also be unexpectedly large in the cold. Indeed, voluntary fluid restriction behaviors have been reported as a means to combat void frequency resulting from cold-diuresis, and heavy exercise can still produce copious sweating in a cold environment (118). When fever and sweating accompany diarrhea, the type of water loss with gastrointestinal illness can also be mixed (165, 258). The presence of both dehydration types could present a significant diagnostic challenge (234, 258)

since neither osmotic nor volume compensation thresholds might be reached. Under these circumstances it may be particularly difficult to distinguish between a hypo-osmotic loss of body water (intracellular dehydration) and an iso-osmotic loss of body water (extracellular dehydration).

Dehydration Assessment

Basic and advanced principles

Table 1 offers a list of 10 basic questions one should consider when selecting any dehydration assessment measure *a priori*. The top half of the table lists questions that are frequently asked and answered in the literature (14, 59, 241), while the bottom half are very infrequently considered. Dehydration assessment methods discussed in this review will touch upon the frequent questions where appropriate, but will focus on the sorely needed answers to infrequently asked questions that offer novel insight related to diagnostic decision making and biological variation. The concept of using diagnostic accuracy toward dehydration assessment is well established in clinical medicine (243) but less so in sports medicine (22, 54, 242). The same is true of biological variation analysis (53–55, 106, 112).

The degree to which any assessment method can correctly classify an individual as dehydrated using a single measurement (cut point) can be described by its diagnostic quality or accuracy (385). Diagnostic accuracy includes both test sensitivity (positive when dehydration is present) and specificity (negative when dehydration is absent). Although high diagnostic accuracy necessarily includes good sensitivity and specificity, a test may still be useful with only high sensitivity or specificity, depending on the purpose of the test (liberal or conservative approach to treatment) and the costs/benefits associated with errors in decision making (385). The use of receiver operating characteristic curves to explore diagnostic

accuracy in the evaluation and introduction of test procedures, such as biomarker assessment, is well-defined.

It should not be surprising that when dehydration assessment measures are compared between well hydrated and dehydrated groups, the means often differ significantly ($P < 0.05$) and an accurate diagnostic cut-point can be identified. However, the importance of the effect magnitude is rarely considered and the magnitude of change in individuals often varies considerably among studies despite similar treatments. This is problematic, particularly since dehydration assessment outside of a laboratory would not involve controlled comparisons, nor would cut-points established on small groups necessarily reflect the same diagnostics when applied to the broader population of interest. Furthermore, while good diagnostic accuracy is a prerequisite for any marker to have practical value, a multitude of factors can produce very different diagnostic cut-points even when study populations and methods are similar (22, 54, 243, 351). For example, diagnostic accuracy $>90\%$ has been reported for drastically different urine osmolality (Uosm) cut-points of 586 (22) and 831 mmol/kg (54) when studying intracellular dehydration in very similar subjects and using nearly identical methods. Thresholds determined using other analytical techniques have ranged from 716 (327) to 1052 mmol/kg (17). There is a strong likelihood that all of these values are correct in context. However, the fact that they differ so widely intuitively negates the diagnostic usefulness of applying any single Uosm cut-point to assess dehydration in a larger population. Similarly, common markers of extracellular dehydration (nonblood loss) generally demonstrate poor diagnostic accuracy (243). These limitations can be quantitatively understood and explained using biological variation analysis.

The use of biological variation analysis as an essential prerequisite to the evaluation and introduction of test procedures is, like diagnostic accuracy, also well defined (111, 112). Biological variation analysis informs strategies for setting population-based and subject-specific reference values, which themselves are critical to diagnostic accuracy testing of groups in many clinical settings (112). Variation exists both within (i.e., CVw) and between (CVb) people, but the variance is typically greater between (112, 161). When $CVb > CVw$, the dispersion of values for any individual will span only a small part of the typical population reference interval. Even highly unusual changes for an individual will likely go undetected as a consequence, thus limiting the efficacy of diagnosis and screening from single test measurements or cut-points (112, 161). According to Harris (161), the ratio CVw/CVb , known as the Individuality Index (*I* Index), informs us about the trustworthiness of labeling a single value or cut point as atypical within the context of the broader population reference interval. When the ratio is <0.60 , individual fluctuations will generally go unnoticed, while atypical values will be captured with greater frequency with values ≥ 1.4 . However, the probability of identifying an atypical individual result within a population-based reference interval increases rapidly as the ratio exceeds 0.60 and approaches 1.0 (161), particularly if a

Table 1 Questions to Ask When Selecting a Dehydration Assessment Measure

Frequently asked questions

- How difficult and expensive is the measurement to perform (practicality)?
- How quickly is a measurement result obtained (seconds, minutes, and hours)?
- In what setting will the measurement be made (clinical and field)?
- How easily is the measurement confounded (require controls)?
- What is the measurement variation (imprecision)?

Infrequently Asked questions

- What type of dehydration is most likely (intracellular, extracellular, or both)?
- What magnitude of dehydration is desired for detection (threshold)?
- Is good measurement sensitivity, specificity, or both desired?
- Is the index of individuality known (within- and between-person variation)?
- Can a reference change value be calculated (serial monitoring)?

Table 2 Biological Variation of Body Fluids: Potential for Spot (*I* Index) and Serial (*H* Index) Dehydration Assessment

	CVw	CVb	<i>I</i> Index ¹	<i>H</i> Index ²	References
Blood					
Posm	0.80-1.9	0.90-1.5	0.87-1.4**	PASS	Jahan et al. (190); Fraser et al. (116); Ricos et al. (292); Cheuvront et al. (54)
PNa+	0.60-0.86	0.60-1.0	0.70-1.0*	PASS	Fraser et al. (116); Ricos et al. (292)
Hct	2.4-2.9	5.2-6.4	0.44-0.50	FAIL	Fraser et al. (116); Dot et al. (88); Nunes et al. (269)
PAVP ³	6.2	29.8	0.21	FAIL	Zerbe et al. (384)
PALD	26-36	28-40	0.73-1.3*	FAIL	Ricos et al. (291); Ahokoski et al. (9)
Urine					
Uosm	17-28	21-58	0.48-.81	FAIL	Shephard et al. (323); Cheuvront et al. (54)
Usg	0.40-0.44	0.61-1.0	0.40-0.72	PASS	Cheuvront et al. (54); Fortes et al. (106)
Ucol	31	47	0.66	FAIL	Cheuvront et al. (54)
Saliva and tears					
Sosm	9.5	36	0.26	FAIL	Cheuvront et al. (54)
Tears					
Tosm	2.2	2.8	0.79*	PASS	Fortes et al. (106)

¹Calculated from CVw/CVb.²PASS requires CVw < 2.8% (i.e., *H* Index < 2.0 calculated for duplicate measures).³Calculated from duplicate measures where $SD = (\sum d^2/2n)^{1/2}$ after adding 1 pg/mL (basal) to the regression coefficient for P_{AVP} in pg/mL per mmol/kg (mean resting $P_{AVP} = 1.8$ pg/mL).

*0.6-1.3 (moderate potential).

**≥ 1.4 (strong potential).

Note: All *I* Index and *H* Index calculations assume that CVa does not inflate physiologic variation (see text for explanation).

confirmatory measurement is made (279, 280). Stratification by age, sex, health, or disease may also be used to improve the index by virtue of shrinking CVb (113). The *I* Index is not particularly large for most measures, including Uosm (Table 2), which helps explain the wide range of proposed diagnostic Uosm cut-points (17, 22, 54, 327).

When the *I* Index is small, serial monitoring (two measurements) is preferred (when possible) since it isolates within-person changes that can then be interpreted in terms of probability. For an individual change to be significant, it must exceed $\sqrt{2} \times Z \times CVw$, where *Z* is the number of standard deviations appropriate for the desired probability (112, 115). In other words, any difference in serial results must be greater than inherent variation for a change to be considered

significant (113-115). The so-called “reference change value” (163) can then be used to guide decision making for an individual. It may even be adopted ubiquitously, with caution, if a true constant variance in CVw can be demonstrated from limited numbers of serial measurements in individuals comprising a larger group (163). The heterogeneity index (*H* Index) provides this insight (160, 162, 163).

The calculations required for the *I* index and *H* Index should ideally include analytical variation (i.e., CVa) as well. A detailed treatment of the CVa applied to the multitude of dehydration assessment methods goes beyond the purpose of this review. Small CVa is highly desirable for many reasons, but will vary among laboratories and equipment (110). In many cases where modern equipment is used and technical

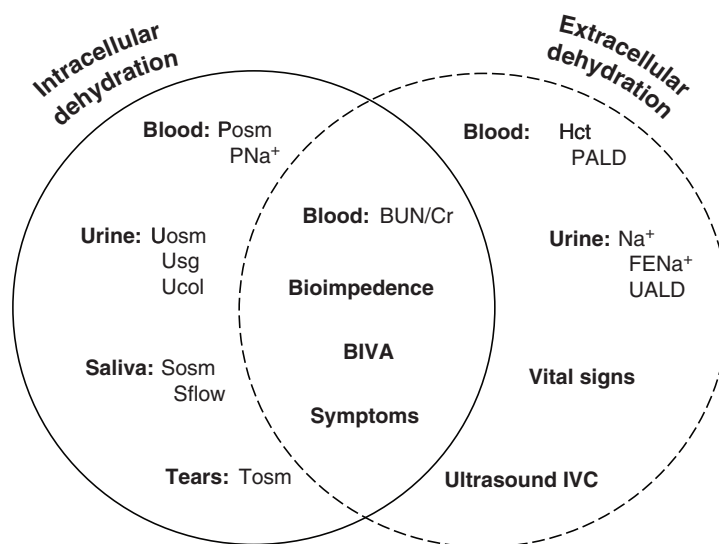


Figure 4 Common dehydration assessment measures grouped on physiologic grounds according to their potential for detecting intracellular dehydration, extracellular dehydration, or both. Within each category, measurements are subdivided by body fluid, technology, or technique.

expertise is high, the CV_a will be less than 0.5CV_w, which inflates the CV_w marginally by ~12% (i.e., CV_w × 1.12) when the variances are properly summed (110). We, therefore, simplify by using CV_w alone in Table 2 to estimate *I* Index and H Index potentials. The principles and practice of biological variation analysis go well beyond what is described here. The interested reader is encouraged to consult Fraser (112) for additional details.

There are many different dehydration assessment methods that could be evaluated using the criteria in Table 1. The majority can be grouped on physiologic grounds into one of three broad categories; those that may detect intracellular dehydration, extracellular dehydration, or both (Fig. 4). Within each category, methods can be subdivided by body fluid, technology, or technique. Although the assessment methods outlined in Figure 4 may not be all-inclusive, they were selected as a fair representation of the most commonly used and studied in the scientific literature.

Methods for assessing intracellular dehydration

Blood

An increased concentration of the blood has long been considered a good index of the degree of intracellular dehydration of the whole body (237). Many dissolved substances increase reliably in response to blood dessication (anhydremia) (237) and could serve as candidate markers of dehydration. In fact, Adolph observed that the concentration of plasma was inversely proportional to circulating BV and concluded that, except for observed changes in body mass, the concentrating effect of dehydration on plasma was the most accurate measure of intracellular dehydration (6).

Direct measurement of anhydremia by the gravimetric measurement of plasma water can be derived by indirect assessment using osmometric methods (10) or quantitative measurement of sodium using ion selective electrodes (266). The primary uses of Posm and plasma sodium for assessing intracellular dehydration can be understood by the osmotic regulation of body water balance by osmoreceptors (35, 361) and the primary, though not exclusive, contribution that sodium makes to Posm (94). Both measurements have moderate to large nosological sensitivity (*I* Index) for identifying an atypical value within the population reference interval, and both display constant CV_w variance for trustworthy reference change value calculation (Table 2). A favorable *I* Index allows assignment of a “threshold” value (with variance term) for dehydration classification (e.g., 54). A favorable H Index allows determination of the probability that any observed change in concentration reflects the occurrence of dehydration, which can be gauged in both quantitative and qualitative terms (55).

It must be recognized that hypo-osmotic body water losses that result in hyperosmolality do not always result in hypernatremia (57, 58, 77, 187). In an experiment in which intracellular dehydration by 2% and 6% of body mass was produced by copious exercise-sweat losses, the slopes of the lines relating the change in Posm and sodium concentrations with percentage dehydration were significantly different, increasing by 3 units and 1 unit for every unit change (1%) in dehydration for Posm and sodium, respectively (57). The authors suggest that sweat sodium losses may explain the discrepancy and that other ions played a larger relative role as water was lost from the blood. If, on the other hand, elevated penetrating solutes like blood urea nitrogen (BUN) contributed more than

expected (288), then the actual osmotic stress on the cell may have been less. It is clear, nonetheless, that Posm provided a superior assessment of measured TBW losses and also estimated free water losses better than sodium (57). It might seem that elevated AVP measures would also be useful for dehydration assessment given its close association with Posm (294) and its step-wise increase in response to graded dehydration (251). However, AVP is analytically more complex, has a fairly short half-life, it can increase substantially in response to numerous nonosmotic or nonvolume stimuli (251, 257, 322, 361), and both its I and H index are unfavorable (Table 2).

If we accept Posm as the best measure in the blood for assessing intracellular dehydration, we must also acknowledge its limitations. Posm and plasma tonicity (effective osmolality) are similar quantities in health (208), but pathological glucose, BUN, or other osmoles have the potential to increase Posm independent of dehydration. Fortunately, dehydration can be ruled out in these circumstances by calculation of the osmole gap (208). Fluctuations in the volume of body fluid compartments will also affect Posm. For example, consumption of a large meal can increase Posm because of the osmolar shift of water out of the vasculature and into the gut (132). In contrast, simple changes in posture (202, 326) and even low intensity exercise ($\leq 40\%$ $\text{VO}_{2\text{max}}$) (65, 151, 251) produce little effect, probably because osmotic equilibrium is not disrupted by the two-way fluid flow between the ISF and PV spaces which share the same osmotic pressures (62, 151, 267). Higher exercise intensities increase Posm as a result of greater intravascular volume losses (hemoconcentration) (65) and the presence of lactic acid, but Posm recovery appears complete within 20 to 30 min of exercise withdrawal (151).

Urine

To understand the potential for using urine as a dehydration assessment marker, it is important to distinguish between everyday renal activity and renal activity in direct response to dehydration. Under ordinary circumstances, urine can be regarded as containing substances that the body contained in excess (3). Day-to-day osmotic constancy of the blood is maintained by renal water excretion or conservation in response to *ad libitum* food and fluid intakes (281, 322, 330). Small changes in Posm are buffered by large fluctuations in urine osmolality (Uosm) via exquisite renal sensitivity to AVP, whereby a ~ 1 mmol/kg increase in Posm will increase Uosm by ~ 90 mmol/kg (294). As a consequence, under conditions of ordinary daily body water flux (input – output), osmotic constancy of the blood ($\sim 1\%$ or 3 mmol/kg) would be expected to fluctuate in accordance with ~ 270 mmol/kg fluctuations in Uosm, similar to the measured day-to-day CVw in Uosm (54). As long as fluid output does not exceed fluid input, normal renal function ensures that Posm is well maintained even when fluid intakes vary widely (16, 278). In fact, the capacity for the kidney to dilute or concentrate urine covers a very large range (Fig. 5) (125), depending on the

balance between fluid intakes, losses, renal solute load (diet and metabolic), and renal blood flow.

The measurement of 24 h urine volume can provide information about minimal fluid intake adequacy (125, 212), but it is a highly impractical method for use in dehydration assessment. Like blood, the most commonly used surrogate of urine volume is urine concentration, which varies inversely with volume. However, concentrated urine can be the result of a reduced urine water fraction (dehydration) or solute addition. For example, typical 1.5 L/day urine outputs double in concentration (e.g., 450–900 mmol/kg) with a doubling of the renal solute load (e.g., 600–1200 mmol/day) (Fig. 5) (125). The addition of solute in the form of antecedent diet, catabolic by-products of protein metabolism (illness and exercise), or the effects of exercise on reducing renal blood flow may all, therefore, increase urine concentration measures, thus mimicking the extraction of water (159, 196, 226, 247, 277, 287). All urine concentration measures are also subject to timing and uniformity concerns that manifest empirically as differences between first morning, 24 h, and spot urine measures (145, 323). Without exception, urine concentration thresholds often used to denote a well-hydrated state are first morning measurements (17, 22, 54, 159, 327). First morning measurements are more uniform as they are generally immune to acute alterations in diet, fluid intake, or activity (196, 226, 247, 277, 286, 287). It has also been demonstrated that concentration of first morning urine reflects 24-h concentrations (145). A troubling trend in the sports medicine literature is to reference the concentration of a spot urine sample in athletes to first morning thresholds in an attempt to classify a person's hydration state in real time. Spot urine samples are not interchangeable with first morning urine samples (145, 323). In fact, Gowans et al. (145) demonstrated that spot urine concentrations differed from 24-h urine concentrations by >2.77 SD in 77/86 individuals who they tested. It is, therefore, suggested that this practice be curtailed.

In response to intracellular dehydration, the discriminatory power of renal excretion measures should always be secondary to changes in Posm (58, 324, 329). It stands repeating that an increase in Posm produces cell shrinkage due to the transmembrane flow of water out of the cell. Cell shrinkage stimulates neuronal firing of osmoreceptors, the hypothalamic release of AVP, and the production of concentrated urine via renal water reabsorption (12, 35, 361). Demonstrable (high probability) changes in Uosm should therefore be a reflection of demonstrable (high probability) changes in Posm. Without blood this would be very difficult to ascertain from a single urine measurement, particularly since urine concentrations can be easily influenced by factors other than dehydration. However, when first morning measurements are possible, most measures of urine concentration (Uosm, urine specific gravity, and urine color) increase and provide good diagnostic accuracy for diagnosing intracellular dehydration under controlled laboratory conditions (54). Not unexpectedly, however, neither Uosm, urine-specific gravity, nor color displays a favorable I Index (Table 2), suggesting that conventional

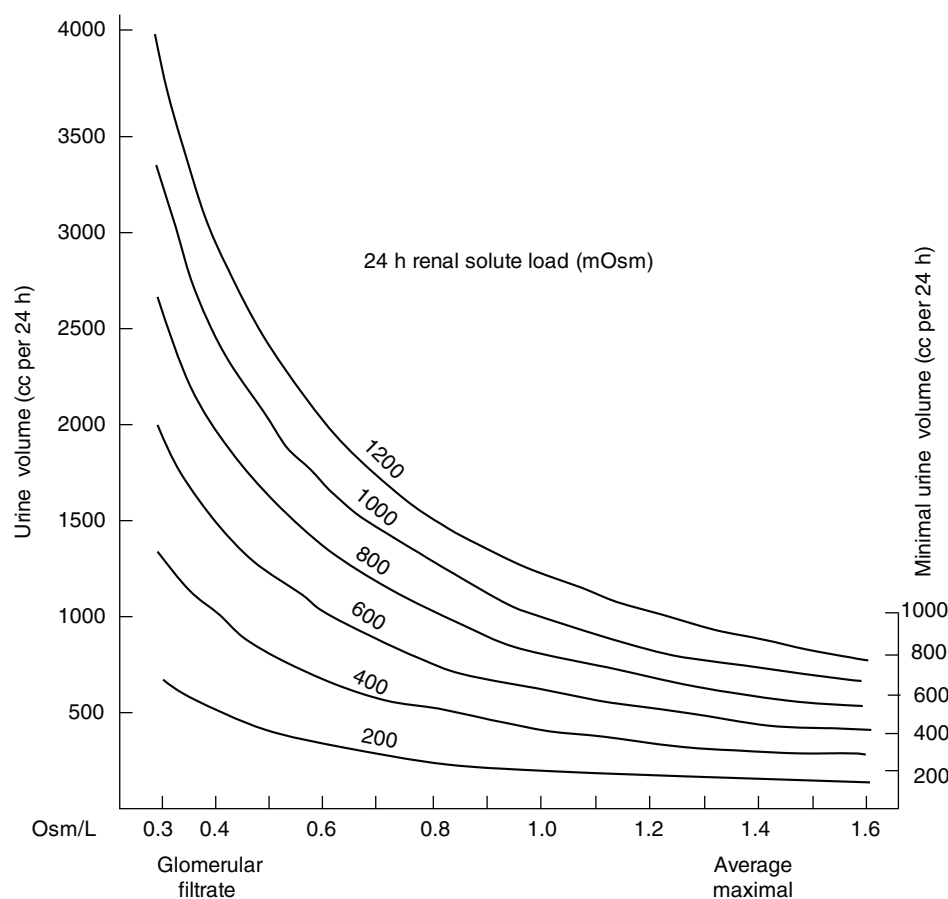


Figure 5 Inter-relationship between urine volume, concentration, and renal solute load. The logical interpretation of dehydration from concentrated urine is based on the reciprocal nature of volume and concentration, but can be confounded by renal solute load. Redrawn, from Gamble and Butler (125) with permission from Wiley.

reference ranges will be insensitive to potentially important within-person changes in urine concentrations (161, 323). Only urine specific gravity appears useful for calculating a reference change value (54,55, 106). However, there are many clinical situations in which urine specific gravity is not interchangeable with Uosm (46, 186), thus the former may be best used as a screening tool for low risk situations.

Another important way in which both Posm and Uosm may be used is in establishing a well-hydrated baseline. In 1957, Smith (334) recognized the baseline problem as a technical research difficulty when studying diuretic and antidiuretic phenomena. He suggested a Uosm/Posm (U/P) ratio of 1.0 where one would be in exact water balance. It has more recently been suggested that true water balance may be better expressed as a matching between plasma and urinary electrolyte concentrations since a considerably larger portion of Uosm is urea (~40%) compared with Posm (~1%). As a consequence, any U/P ratio ≤ 1.5 is more consistent with electrolyte free renal water clearance and a water-loaded state (325). Thus, a day-to-day change in body mass of ~1% coupled with a U/P near ~1.5 (or plasma and urine electrolyte equivalence) may be considered an ideal baseline from which to study dehydration.

Saliva and tears

It should not be surprising that other body fluids besides blood and urine concentrate in response to hypo-osmotic body water losses. Adolph and Wills (8) may have been the first to suggest that the absence of salivation could be used as an index of severe dehydration, which they defined as 8% of body mass. In their experiments, the flow of saliva in response to chewing paraffin was inversely associated with dehydration across a broad range, but individual flow rates were not consistently reduced until 4% dehydration was reached (8).

Walsh and colleagues have systematically examined how intracellular dehydration due to exercise-heat stress (365, 366) or fluid restriction (273) affects the flow of saliva and saliva osmolality. In general, both measures perform as expected in response to graded dehydration. Mean values for a group decrease (flow) or increase (osmolality) with statistical significance, but variation remains large among individuals. Diagnostic analyses using saliva osmolality give different cut-points and different diagnostic accuracy values, possibly due to dissimilar baselines or perhaps differences in study methodologies (54, 351). The only examination of biological variation in saliva osmolality (54) revealed highly unfavorable

I and H Index values, suggesting little potential for saliva as a dehydration assessment marker (Table 2). It should also be noted that saliva is subject to practical use issues related to simple oral artifacts and that it is also affected by anything that affects salivation (flow), which includes a multitude of factors. For example, cholinergic influences on salivation may confound interpretations, as pilocarpine has been demonstrated to completely reverse the drop in the flow of saliva during dehydration (8). Simpler scenarios like a 10-s mouth rinse with water also results in subsequent saliva osmolality values that look well hydrated for as long as 15 min despite significant whole body dehydration (99).

The potential for using ocular measures for dehydration assessment is a very recent undertaking. The rationale for this approach is similar to saliva in that, like salivary glands, any reduction in the output of moisture producing glands within the ocular anatomy may be quantitatively measured. Like the absence of salivary flow in relation to severe dehydration in adults (8), the absence of tears in crying children (<3 years old) is among the top four clinical predictors of severe pediatric dehydration (120). When several common ocular measures used for other purposes in ophthalmic medicine were reviewed by Sollanek et al. (336) for their potential use in dehydration assessment, it was concluded that tear osmolality (mmol/L) possessed the most merit. Although it remains unknown whether tear fluid represents a direct filtrate from plasma (357,358), the range in human tear osmolality appears similar to Posm (218) and the two are positively correlated when measured simultaneously (106). With the advent of improved technologies for measuring tear osmolality in the 21st century, Fortes et al. (106) demonstrated that tear osmolality tracks dehydration and Posm incrementally. In addition, decreases in plasma osmolality during rehydration and recovery were also reflected in decreases in tear osmolality, providing confidence that the changes in tear osmolality reflect changes in hydration, rather than an exercise artifact. Tear osmolality may be especially appealing when change values can be measured (Table 2). It may be particularly attractive for use by clinicians because it is less-invasive compared with Posm, requires little expertise to perform, and provides a rapid reading (106). It is important to remark that the true potential for using tear osmolality is difficult to judge from a single study. Furthermore, obvious measurement confounders that might affect tearing or drying of the eyes, such as outdoor exercise (movement convection) and environment (wind, sun, and rain), may limit applications within sports, wilderness, and military medicine.

Methods for assessing extracellular dehydration

Blood and urine

The iso-osmotic water loss that occurs during extracellular dehydration produces little change in Posm, but large losses in ECF volume (23, 234, 258). The viscosity of blood does change, however, as blood is a non-Newtonian fluid with a

viscosity that is proportional to hematocrit (60). In fact, when hematocrit is measured in combination with hemoglobin to correct for potential changes in cell volume (164), it can be used to calculate changes in PV (84). This would seem to make it a good candidate for dehydration assessment, but as a single measurement the I Index is too small and the H Index too large (Table 2). Perhaps more importantly, simple postural changes can produce a ~4 unit change in hematocrit (10% of 40) (157, 353, 356), which itself is two to four times larger than inherent variation. Exercise also reduces PV acutely in proportion to exercise intensity (65). In response to intracellular dehydration by 4% body mass, hematocrit increases by only ~2 units (~5% of 40) (70). This effect may be due in part to a small amount of cell volume shrinkage associated with increased osmotic pressures (148, 372), but is more likely because the TBW deficit is shared between ICF and ECF spaces. A much larger increase in hematocrit is seen with extracellular dehydration. Nadel et al. (260) measured a 7 unit increase in hematocrit (~18% of 40) after extracellular dehydration to just 2.7% body mass using two different diuretics ingested over 4 days. Hematocrit may therefore serve as a sensitive measure for serial assessment of extracellular dehydration under conditions where posture, body movement, and changes in skin temperature (164) are carefully controlled. However, reference change values for serial monitoring cannot be generalized with confidence (Table 2). The wide range of typical values, even when narrowed by sex-stratification (112, 207), combined with diseases and medical conditions that commonly alter hematocrit would seem to agree with its limited potential as a spot measurement (Table 2).

Large increases in hematocrit are generally consistent with large decreases in ECF volume. The slower of two compensatory mechanisms that acts to protect circulatory volume involves fluid regulatory hormones (317). Water and electrolyte losses that produce volume depletion of magnitude >10% BV would be expected to stimulate AVP and aldosterone production to retain water and sodium at the kidney (Fig. 3) (23, 24, 91, 216, 221, 304). Although this is not always the case (135), it has been suggested that renal excretion of sodium and water may provide the best diagnostic indication of an adequate arterial circulation (317). Indeed, an increase in urinary aldosterone accompanying a decrease in urinary sodium (<20 mEq/L), or a low fractional excretion of sodium (<1% FE_{Na}) may be considered indicators of volume depletion (324, 344). However, they may also be indicators of renal dysfunction or dietary sodium inadequacy (225, 324, 344). Caution is recommended when interpreting urinary sodium measurements (324, 344) and FE_{Na} requires both blood and urine measures of sodium and creatinine in its calculation. Plasma aldosterone is not a candidate for serial monitoring of extracellular dehydration, though it does appear to have potential as a spot marker (Table 2). But like AVP, aldosterone measurement also requires analytical complexity and has a fairly short half-life. Once again, the choice of dehydration assessment measure must be an informed one (Table 1).

Vital signs

The faster of two compensatory mechanisms that act to protect circulatory volume during extracellular dehydration is the neuro-humoral responses to baroreceptor stimulation, including increased renal and peripheral vascular resistance (317, 378). A reduction in renal blood flow (and glomerular filtration rate, GFR) precedes and complements fluid regulatory hormone retention of renal water and sodium (317). Peripheral vascular effects on the circulation are typically gauged using vital signs that include heart rate, blood pressure, or various indices of the same. In general, gross blood pressure measures (rest or postural) have poor diagnostic potential for dehydration assessment (243) even in response to severe hemodynamic challenges (67, 205, 222). Physiologic defense of blood pressure is robust, redundant, and highly variable (47, 67, 364). For example, the use of lower body negative pressure to a level consistent with a 0.5 to 1.0 L blood loss (67) results in almost no discernible change in mean arterial pressure (MAP) or pulse pressure (64). Stroke volume declines, but muscle sympathetic nerve activity compensates (64). In fact, MAP is maintained with simulated blood losses >1.0 L, though pulse pressure begins to decline noticeably (>5 mmHg) when simulated blood losses exceed ~0.5 L (64). The nonosmotic relationship between AVP and MAP requires >10% decline in MAP for AVP to rise above resting values (25), which probably explains the absence of an AVP response to actual mild hemorrhage (~0.5 L) (135). Recall that PV does not decline by ~0.5 L (~10% BV) during whole body extracellular dehydration until a 4% loss of body mass is achieved (~3.0 L TBW loss). The same PV loss in intracellular dehydration would require ~7% whole body dehydration (~5.0 L TBW loss) (Fig. 1) (56, 58).

A reduction in PV decreases venous return to the heart and is reflected in a smaller cardiac stroke volume (119). Movement from sitting to standing also decreases stroke volume by increasing venous pooling in dependent limbs. When venous return is reduced, a rapid increase in heart rate and peripheral vascular resistance is required to maintain MAP (378). It is therefore logical that a greater change in heart rate should occur upon standing when dehydrated (42), particularly with extracellular dehydration (299) and especially when the magnitude of dehydration is severe. However, the heart rate response to orthostasis appears highly variable and largely insensitive to dehydration (27, 119, 185, 191, 205, 222). Possible explanations include, but are not limited to, variability among orthostatic test stressors (185), large CVw and/or CVb in baroreflex responsiveness (66, 92, 128), or differences in the type and magnitude of dehydration tested (53).

The change in heart rate from sitting-to-standing (SSΔHR) is commonly recommended as one of several gross screening tests to aid in the detection of dehydration (222, 235, 243, 302, 378, 381), particularly in ambulatory triage situations (235, 381). Cheuvront and colleagues (53) compared the SSΔHR response to moderate intracellular and

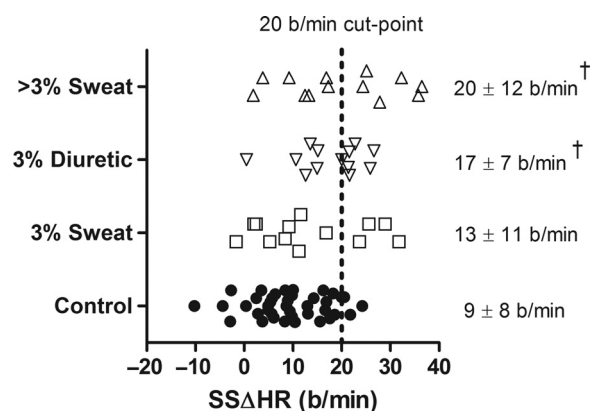


Figure 6 Comparison of 1-min SSΔHR during control (well-hydrated) and after matching dehydration by furosemide or sweat loss to 3% of body mass, and after 4.5% dehydration using sweat loss. Values are mean ± SD. † Significantly different from control (well-hydrated) ($P < 0.05$). 20 b/min cut-point provided 67% diagnostic accuracy for dehydration. Adapted, with permission, from Cheuvront et al. (53).

extracellular dehydration (~3% body mass), as well as more severe (>3% body mass) intracellular dehydration. Variability in sitting and standing heart rates were very large. Application of the 20 b/min cut-point suggested by Witting and Gallagher (381) resulted in excellent diagnostic specificity (90%), but poor sensitivity (44%) and weak overall accuracy (67%) (53) (Fig. 6). The same diagnostic accuracy but converse of specificity and sensitivity could be achieved with an 11 b/min cut-point. Thus, SSΔHR exhibited unfavorable biological variation and did not afford good diagnostic accuracy, but could theoretically be a useful test if a liberal (20 b/min) or conservative (11 b/min) treatment approach was desired (Table 1). One important assumption underlying this conclusion is that test conditions are carefully controlled and free from any influence of independent factors known to influence heart rate.

Ultrasound inferior vena cava diameter

Sonographic measurement of the inferior vena cava (IVC) diameter is an attractive, noninvasive alternative to invasive central venous pressure (CVP) measures and for assessing intravascular volume status (340), which would be reduced in response to extracellular dehydration. Arguments for using ultrasound IVC include technological advances that have made point-of-care ultrasonography a safe and effective bedside tool (255), the IVC does not appear to be influenced by the neuro-humoral response to volume losses (263), and in properly trained hands it can be a highly repeatable measurement (105). Dipti et al. (85) meta-analyzed five studies (three case control and two before-after) whereby 86 cases of volume loss were compared to 189 control or volume replete cases. It is worth noting that although visual estimation of volume status is sometimes used by sonographers for classification (105), all cases reviewed by Dipti et al. (85) were in various stages of diagnosed shock. The grand mean IVC

diameter in volume depleted cases was 6.3 mm less than controls. This is similar to the 5 mm reduction in IVC reported by Lyon et al. (228) in response to 0.45 L phlebotomy. Katzarski et al. (197) also reported that IVC diameter tracked acute volume loss and regain during dialysis. However, differences among the mean volume depletion cases reviewed by Dipti et al. (85) were as large as 10 mm, while differences among controls were as large as 18 mm (85). The use of IVC in calculating a collapsibility index ($IVC-CI = [IVC \text{ expiration} - IVC \text{ inspiration}] / IVC \text{ expiration}$) may improve dehydration assessment (340, 375), particularly in extreme cases (340). Although no objective study of IVC biological variation has been performed, it would appear from the variation between people that the greatest potential for using ultrasound IVC for detecting extracellular dehydration would be only for serial measures in the same person (375). It is also worth noting that since IVC is a surrogate measure for CVP, any inaccuracy for using CVP to detect volume depletion (236) will likewise be manifest when using IVC.

Methods for assessing mixed dehydration

Blood

The ratio of BUN to creatinine (BUN/Cr) is significant in the analysis of renal function (87, 179). The relative constancy of creatinine excretion, coupled with the variable balance between excretion and reabsorption of BUN, give a typical BUN/Cr ratio of between 10 and 20:1 (87, 179). BUN reabsorption is an inverse function of urine flow rate and is mediated by AVP (307), thus BUN/Cr may be used for assessing dehydration stimulated by either osmotic or volume stimuli. This potential, however, seems very limited by the many factors which can increase BUN (production) independent of AVP-stimulated reabsorption, including protein intake and catabolic states (87, 179, 296). The inverse relationship between BUN reabsorption and urine flow is actually parabolic (179), thus it is unclear what magnitude of dehydration would be required to reduce GFR enough to exceed the typical upper BUN/Cr limit of 20:1. In one study of children ages 2 weeks to 12 years, BUN/Cr was significantly associated with dehydration due to gastroenteritis, but BUN/Cr ratios from 20 to 40:1 provided <70% diagnostic accuracy for dehydration >5% body mass (352). For these reasons, it is recommended that BUN/Cr be used cautiously for dehydration assessment (296). It is also worth noting that where there is interest in dehydration assessment using serial changes or spot measures, the covariance between any two measures that make up a ratio, such as BUN/Cr, may increase false negative or false positive results, depending on whether the numerator or denominator is confounded (376).

Bioimpedance

Body fluids are good conductors of electrical current. Bioelectrical impedance (Z) to the flow of electrical current can

be related to volume [$V = \rho(L^2/R)$] (227), where L^2 is body length measured by convenience as the square of standing height (Ht^2) and Z is predicted by resistance (R). The resistivity of body fluids (ρ) must be measured or assumed (130). The impedance quotient (Ht^2/R) can be regressed against V (i.e., TBW) to produce population-specific slope (m) and intercept terms (c) which then form the basis of predicting TBW using regression: $TBW = mHt^2/R + c$ (170). Single, low frequency currents (e.g., 50 kHz) do not penetrate cells and thus provide an estimate of TBW based on the measurement of ECF and the strong linear dependence between TBW and ECF (321). The partitioning of ICF and ECF spaces is made possible by the addition of high-frequency currents which do penetrate cells (TBW), thus ICF is calculated from the difference between estimates of TBW (high frequency) and ECF (low frequency). Multi-frequency bioimpedance spectroscopy (BIS) affords compartmental measurements by fitting multi-frequency impedance measures to a biophysical Cole-Cole plot (68, 79, 359). A wide variety of different single and multi-frequency bioimpedance models have been published.

Good agreement has been reported between single frequency impedance measures and TBW by dilution. Similarly, BIS provides good correspondence to gold standard dilution methods for TBW, ECF, and by deduction ICF volumes in healthy and well-hydrated individuals. However, despite small mean differences and large correlation coefficients, the error variance is always large (13, 98, 170) and well in excess of 2 L for TBW (or ~5% of TBW). For individuals, 95% limits of agreement would exceed 4 L for TBW. Indeed, the clinical validity for using bioelectrical impedance models has been questioned on the basis that limits of agreement are too wide for accurate application at the individual level (36). Importantly, the large variance in the differences between dilution and TBW estimates have been achieved while adhering to strict measurement methodology (97, 210), which underscores the limited practical application of impedance for many relevant treatment settings where measurement conditions cannot be precisely controlled (36, 210, 271).

Neither single nor multi-frequency (6-500 kHz) impedance methods appear useful for detecting body water losses in humans, even when measurement conditions are highly standardized (36, 271). A partial explanation might rest with the empirical derivation of model-specific constants from a well-hydrated population with a typical ICF:ECF ratio (36, 68, 271). Additional explanations may rest with limitations that are unique to a given set of model parameters and assumptions (153). Even the most sophisticated of bioimpedance methods, such as the $0/\infty$ -kHz parallel Cole-Cole and Hanai mixture models (79) contain potential errors (36, 369). For example, Gudivaka et al. (153) reported that only one of seven models (i.e., $0/\infty$ -kHz parallel Cole-Cole) performed accurately (on average) when the ICF:ECF ratio was experimentally altered despite heroic care in reducing propagated errors. Theoretical models remain limited by the unknown effects of body water loss on resistivity

constants for ICF and ECF (ρ_{ICF} , ρ_{ECF}) (369). For example, an iso-osmotic TBW loss of ~ 2 L can reduce PV enough to raise hematocrit by 7 points (260). In theory, this would increase ρ_{ECF} and produce a surfeit ECF prediction (130, 369).

BIVA

A simplified twist on the use of single frequency bioimpedance for detecting body water losses involves bioimpedance vector analysis (BIVA), or the RXc graph (283). A single low-frequency current (50 kHz) is used to measure “whole body” R and reactance (Xc), which are then standardized for height (i.e., R/H and Xc/H). After measuring R/H and Xc/H in a reference population, a plot of R/H (x -axis) and Xc/H (y -axis) is created whereby 50%, 75%, and 95% tolerance ellipses are drawn assuming a bivariate normal distribution for R/H and Xc/H (283). Direct impedance measurements of R and Xc can then be plotted on the graph so that the position of an individual vector can be classified probabilistically with respect to its position within the reference population. Because each RXc graph is created unique to the impedance analyzer used, an RXc score graph was created to normalize vectors to Z scores (i.e., difference in individual R and X values from the mean/SD) (282). Dimensionless $Z(R)$ and $Z(X)$ scores can then be plotted with tolerance ellipses preserved.

A proposed strength of the RXc graph method is that repeated measurements on the same subject allow within-person changes in TBW to be compared relative to between-person norms (ellipses) (282). Any point displaced outside the 95% ellipse in quadrant I of a Cartesian coordinate system, whereby all x/y ratios are positive, is consistent with below normal body water. Serial measures from any point in quadrant I, whereby lines extend away from or toward the origin, indicate relative dehydration or rehydration, respectively (282) (Fig. 7). It is uncertain how well the RXc score graph can identify whole body water losses that result in intracellular or extracellular dehydration. It has been demonstrated that the RXc graph captures the loss of ultrafiltered volume in dialysis patients better than single frequency bioimpedance alone (154). Any change in the RXc vector should be a function of a change in ECF only (321), since a 50 kHz current does not penetrate cells to capture TBW (79). The RXc graph might therefore find use as a serial monitoring tool, but it should be remembered that simpler measures, such as changes in body mass, provide nearly perfect correspondence to ultrafiltered fluid volumes (ECF) during similar dialysis experiments (71, 189). A criticism of the RXc graph is that it is subject to the same errors and uncertainties as all single frequency impedance measures. For practical purposes, any desire to back-calculate absolute volume losses (and prescribe corrections) would be limited by the requirement to use regression equations formulated from normally hydrated persons, thus negating the attraction of this approach for quantifying dehydration (354).

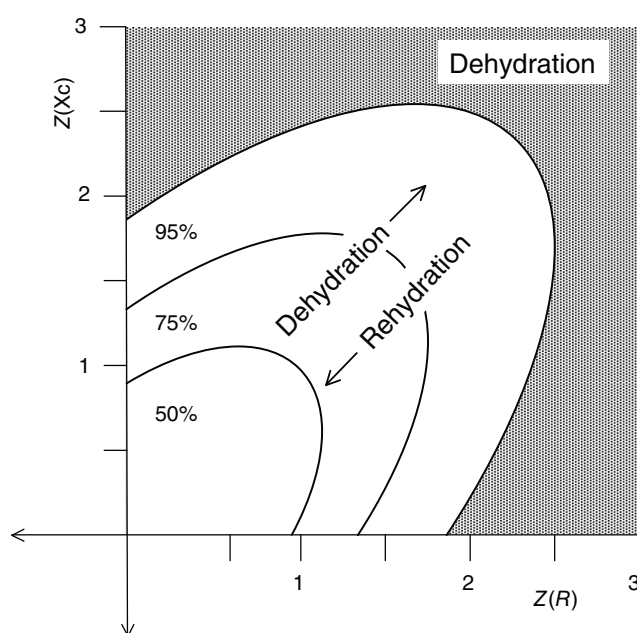


Figure 7 Quadrant I of a typical RXc score graph. Axes represent dimensionless $Z(X)$ and $Z(R)$ scores plotted with population tolerance ellipses. Any point displaced outside the 95% ellipse in quadrant I is consistent with dehydration. Serial measures from any point in quadrant I extending away from or toward the origin indicate relative dehydration or rehydration, respectively (282).

Symptoms

At the turn of the 20th century, McGee (244) accompanied scientists across the Arizona southwest and wrote vividly of the progression of physical symptoms experienced by surveyors during prolonged exposure to the desert sun. McGee used a qualitative five stage classification system to describe the severity of symptoms, to which Adolph added additional measurement details in association with quantitative levels of dehydration (% body mass) around WWII (7). The number of different symptoms is numerous and little is to be gained by reviewing the plausible merit of each. In general, physical symptoms at the low end of the dehydration spectrum are nonspecific; they are common among many different medical maladies (e.g., headache, apathy, and fatigue). Individual physical symptoms such as sunken eyes, capillary refill time, and skin turgor have proven diagnostically inaccurate in a variety of dehydrated adult populations (242, 243, 363). Ladell (212), considering his own observations between field and laboratory, concluded that while the underlying physiology was the same, dehydration by exposure to the elements (desert and lifeboat) more often produced distinguishing gross symptoms than the same level of laboratory dehydration. Excess adrenal activity associated with genuine survival anxiety was one explanation offered (212). It is also clear that the levels of dehydration observed by McGee and Adolph were well in excess of those typically produced in laboratories or estimated in emergency room settings. However, by the time these symptoms become obvious dehydration is almost certainly detectable using any standard test.

Physical symptoms are frequently used singly or in combination when assessing pediatric dehydration, but debate remains over whether laboratory measures or physical symptoms provide the best diagnostic criteria (120, 143, 165, 342, 343, 352, 360). Recall that the type of dehydration is critical to the potential validity of any assessment measurement. Just as it is critical to laboratory tests in adults (58), it is no less critical when assessing physical symptoms in children. Recall also that the absolute magnitude of dehydration matters (58). An excellent example which addresses both of these issues was published by Laron in 1957 (214). The potential for using skin turgor to assess dehydration was tested in 25 children ages 1 month to 3.5 years and one child age 9. Dehydration due to diarrhea (extracellular) and water deprivation (intracellular) were compared. The slopes of the lines relating skin turgor to dehydration were similar between groups, but the intercept term differed substantially (214). There was no loss of skin elasticity in response to water deprivation until dehydration exceeded 6% of body mass, while only 2.5% dehydration was required with diarrhea. Extracellular dehydration required roughly half the level of TBW loss because ECF volume loss is double that of intracellular dehydration (Fig. 1) (56). The preferential loss of skin ISF, due to the lower oxygen requirement of skin, likely explains the change in skin turgor (173, 214). It is also likely that physical symptoms may be more pronounced in the very young who possess limited physiologic reserves (238).

The most widely studied of dehydration symptoms is thirst. McGee's account of 1906, titled "Desert Thirst as Disease" (244), recognized three different types of thirst to include ordinary, sea, and desert. His experience in the desert allowed him to further classify desert thirst into five stages, the first of which is akin to ordinary physiologic thirst. As mentioned previously, osmoreceptor-mediated thirst is independent of AVP and stimulated by direct neural signaling (35, 80, 81, 246). In fact, the Posm threshold for unequivocal thirst is ~ 10 mmol/kg higher than the Posm threshold for AVP, though there is considerable variation among people (26, 295, 384). Angiotensin II also appears to mediate osmotically driven thirst and may participate in the secondary thirst response that is volume-mediated (11, 12, 35, 246). Thus, although thirst can be triggered by both osmotic and volume stimuli, the osmotic threshold for unequivocal thirst would occur earlier in response to intracellular dehydration ($\sim 2\%$ body mass) compared to extracellular dehydration ($\sim 4\%$ body mass) (58). In fact, thirst may not appear during extracellular dehydration until TBW losses exceed $\sim 6\%$ body mass (258).

Spontaneous fluid intake is governed primarily by feeding behavior as opposed to genuine thirst (78). When fluid is restricted, humans gauge thirst in a linear fashion with TBW loss and Posm (328). However, thirst mechanisms are subject to numerous influences unrelated to body water balance (149). In humans, fluid losses (sweat, vomiting, and diarrhea) can easily outpace oral intakes. Peripheral osmoreceptors and oropharyngeal cues trigger thirst satiety well before volume is fully restored (35, 80, 81, 131, 297, 349, 355) even when

dehydration is substantial (150). This transitory response acts to buffer the presystemic impact of ingested fluids (35) but often leads to involuntary dehydration (149, 301).

Summary and Future Directions

One premise of this review is that the type and magnitude of dehydration are arguably the most essential and underappreciated aspects of dehydration assessment. The potential merit of any candidate marker of dehydration therefore rests upon careful consideration of these points, as well as others discussed herein (Table 1). It is critical that future dehydration assessment work consider the application of diagnostic and biological variation analyses. When properly applied, it will refocus the way in which assessment methods have been interpreted using group trends to the more useful assessment of individuals (111, 113–115). Another future direction in the study of dehydration assessment might be to recognize that cell signaling is at the heart of organism-based homeostatic body water regulation – whether osmotic or volume-mediated. Burg et al. (38), and others (168, 183, 213), have already identified numerous molecular pathways that comprise the cellular response to dehydration. Although this information has not yet been used for the purposes of dehydration assessment, there is great potential. The challenge will be to: (1) validate a convenient mammalian cell model(s) for study and (2) identify and select the best cell biomarkers for measuring relevant magnitudes of intracellular and extracellular dehydration.

Performance

Introduction

The concept of human "performance" is multifaceted and has evolved from early human history to the present day. The relationship between dehydration and work "performance" has likely been apparent for centuries. Early man recognized the importance of hydration to the success of foraging and hunting expeditions, devised various methods to carry water, and planned for water availability during long trips. The importance of water availability to "performance" was also recognized by early militaries (136). Potable water has helped shape military strategy and has even been leveraged as a tactical weapon. Indeed, it has altered the outcome of numerous battles from the earliest recorded history to the present day (136, 250). However, it was not until the early 1900s that the relationship between dehydration and work performance was studied and quantified. These studies were driven not only by concerns for health, but also the desire to analyze the performance of industrial workers to determine the best conditions and treatment for maximum productivity (362).

As previously discussed, day-to-day variation in body mass, or TBW, represents typical physiologic and behavioral body water regulation (147) and is $\leq 1\%$ when fluid intake and activity are controlled (50, 54). As such, day to day TBW

deficits must exceed 1% and approach 2% of body mass to be deemed atypical (58). This point is especially poignant when reviewing studies that propose what is termed “mild” dehydration (<2% body mass loss), especially if a well-hydrated baseline is not first established. A good baseline is typically determined via consecutive daily measures of first morning, postvoid and nude body mass, usually after providing fluid (1–2 L) the evening prior (50, 53, 54, 100, 320). If a well-hydrated baseline body mass is not established, it is unclear what degree of dehydration has been achieved by any acute perturbation. Interestingly, there is an overall consensus in the literature that dehydration of $\geq 2\%$ body mass represents a threshold at which aerobic exercise performance or endurance becomes impaired (308, 309, 312) and may (194) or may not (49, 188) be the threshold at which anaerobic exercise performance or muscular strength (103, 152) becomes impaired. The following section will review various aspects of “performance,” particularly in relation to physiologic changes in Posm and BV that accompany dehydration. In addition, this section will provide a unique analysis of the related dehydration and performance literature to include plausible mechanisms for impaired performance.

Physiology

Dehydration can impact physiologic processes, such as thermoregulation, by virtue of alterations in Posm and BV. It is not the intent of this paper to provide an extensive review of the broad physiologic impact of dehydration. In brief, when dehydration occurs, PV decreases and Posm increases in proportion to the decrease in TBW (187). Plasma hyperosmolality can delay thermoregulatory cutaneous vasodilation and sweating by elevating thresholds for both (108, 350). Thus, dehydration reduces the sweating rate for any given body core temperature, decreases evaporative heat loss (314), and increases heat storage (107, 108, 261). Further, as a result of a reduction in circulating PV, heart rate increases secondary

to a reduction in stroke volume (140, 306, 337) and perceived exertion increases during work (127). The addition of heat stress in combination with dehydration during exercise creates competition between the central and peripheral circulation for limited BV (259, 303) and further exacerbates the physiologic strain for a given exercise task (249, 252, 314). For those interested in this topic, more far-reaching reviews of these responses are available in the literature and are highly recommended (187, 229, 230).

Endurance, strength/power performance

The preceding cursory review of dehydration physiology can help to form a basis for understanding the role that dehydration plays in performance. Mixed dehydration is likely during extended work and exposure to cold or high-altitude environments. However, when considering the potential impact of dehydration on performance, it is imperative to recognize that the majority of circumstances will involve intracellular dehydration due to sweat losses and fluid restriction or unavailability. The principal physiologic mechanism(s) which explain why performance is (or is not) impaired by intracellular dehydration will be considered for endurance and strength/power activities, cognition, and sport skill.

Figure 8 summarizes that dehydration $\geq 2\%$ body mass consistently impairs endurance exercise performance, but probably not strength or power. Typical qualitative (194) or quantitative meta-analyses (144) involve strict exclusionary criteria which ultimately reduces the number of studies for analysis to a handful of experiments. Although there are good reasons for adopting this approach (239), we took a different and less conservative approach by simply counting laboratory and field studies reporting statistically significant impairments in performance for endurance (1944–2010) and strength or power exercise (1961–2012). Studies were located initially using a thorough electronic search of PubMed. An extensive manual search of reference sections of all articles that were

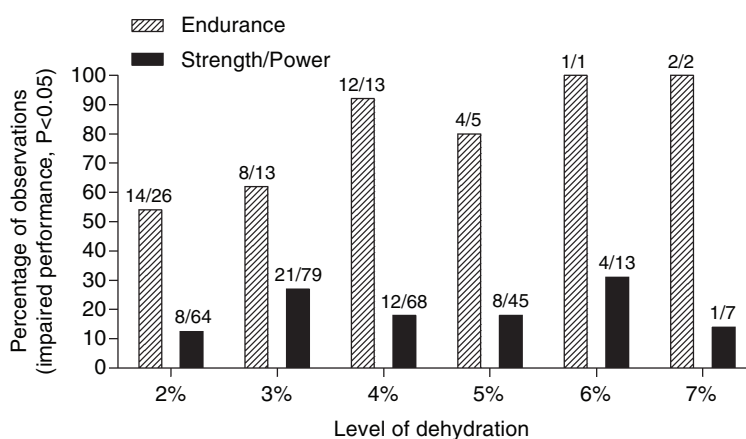


Figure 8 Review of dehydration effects on performance in 34 endurance and 43 strength or power studies. Fractions above bars provide the details for the y-axis percentages. References for the studies reviewed can be found on-line at Supplemental Material.

found during the electronic search, including comprehensive narrative reviews like that of Judelson et al. (194), was also performed. Minimal exclusion criteria were applied, but studies were not included if: (1) dehydration was < 1.5% body mass, (2) the precise level of dehydration was not reported, or (3) substrate was part of the fluid replacement beverage in the control trial (51). The complete set of references for Figure 8 is available online as supplemental material to this text. Dehydration levels were grouped over a range of $\sim 1 \pm 0.5\%$ (e.g., 2% = 1.5–2.4%).

There were 34 endurance exercise studies evaluated which included 60 separate observations. A total of 41/60 observations (68%) were significantly ($P < 0.05$) impaired by dehydration $\geq 2\%$ body mass. For strength and power, 43 studies were examined comprising 276 separate observations. In contrast to endurance exercise, only 54/276 observations (20%) were significantly impaired ($P < 0.05$) by dehydration $\geq 2\%$ body mass. Because laboratory performance studies are often conducted with small numbers of subjects (~ 10 per study or fewer) and may be statistically underpowered to see small, but important effects, we also counted the number of studies reporting a reduction in performance independent of P value with an *a priori* stance that the results should be 50% if due solely to chance. The results were 53/60 negative observations (88%) for endurance exercise of $\geq 2\%$ body mass loss, which further strengthens the conclusions interpreted using statistical significance. For strength and power, the results increased to 177/276 observations (67%), which tends to coincide with findings that strength and power are negatively affected to a small degree (1–3%) (194).

Endurance: Mechanisms of performance impairment

The consequences of dehydration to endurance performance include a shorter exercise time to exhaustion, an obligatory reduction in exercise intensity, or both. Empirical laboratory

and field data support this proposition. Although the size of the dehydration-mediated performance impairment varies greatly among studies, it is commonly two times larger than the typical variation in the measurement itself (51, 56), which illustrates the strong negative effect of dehydration on endurance exercise performance. The preponderance of evidence in Figure 8 is consistent with the consensus statement of the American College of Sports Medicine (309) and the observations of Ladell (211, 212) almost 60 years ago that physiologic functions are preserved until a loss of ~ 2 L of TBW is exceeded, which represents 2% to 3% dehydration for a 70 to 80 kg person. The mechanism(s) by which performance is impaired remains debated (56).

While dehydration can reduce muscle blood flow (137, 138) and alter skeletal muscle metabolism (104) during intense, exhaustive exercise in hot environments, intense exercise in hot environments is limited in duration and unlikely to be affected by dehydration if exercise is begun well hydrated. Headache, or even thirst (312), may also produce subjective feelings of fatigue or loss of vigor, but attempts to measure somatosensory gating of afferent neural signals in association with dehydration and exercise performance are thus far inconclusive (254). Hyperosmolality is an unavoidable consequence of dehydration which helps mediate body core temperature elevation (308) and also contributes to increased cardiovascular strain (72, 249, 380). Dehydration by 2% to 4% body mass also produces an added challenge to meet whole body blood flow requirements by reducing PV up to 10% (Fig. 3), which acts to reduce cardiac filling and stroke volume (127, 138–141, 306). An explanation for a volume-mediated mechanism of impaired endurance performance follows.

Dehydration by 2% to 4% body mass generally reduces $\dot{V}O_{2\max}$ (Table 3). In one study (73) the reduction in $\dot{V}O_{2\max}$ was proportional to the level of dehydration (Table 3). Figure 1 (56) shows how PV loss is proportional to the level of dehydration. A reduction in PV decreases venous return to the heart and is reflected in a smaller cardiac stroke volume

Table 3 Effects of Dehydration and Environment on $\dot{V}O_{2\max}$

Study	Sample size	Environment	Dehydration (% body mass)	$\dot{V}O_{2\max}$ (%)
Nybo et al. (270)	6	15°C perfused suit	–4%	–6%
Caldwell et al. (41)	47	Room temperature	–4%	–3%
Webster et al. (374)	7	Room temperature	–5%	–7%
Buskirk et al. (39)	13	26°C	–5%	–6%*
Ganio et al. (127)	11	30°C	–4%	–9%
Nybo et al. (270)	6	44°C perfused suit	–4%	–16%
Craig & Cummings (73)	9	46°C	–2%	–10%
Craig & Cummings (73)	9	46°C	–4%	–27%

All percentages rounded to the nearest whole integer.

*Buskirk et al. only provided the reduction in $\dot{V}O_{2\max}$ in L (0.2 L). The percentage was estimated assuming a $\dot{V}O_{2\max}$ between 3 and 4 L/min (3.5 L).

(119). Gonzalez-Alonso et al. (140) have clearly shown that stroke volume specifically declines progressively during exercise as dehydration worsens. These facts fit well the observations by Wingo et al. (380) illustrating a proportionality between reductions in stroke volume and reductions in $\text{VO}_{2\text{max}}$. A reduction in $\text{VO}_{2\text{max}}$ when dehydrated would make incremental or constant-rate exercise more difficult to sustain (127, 224, 249) or require a slowing of self-paced exercise to achieve a similar sensation of effort (44, 109, 341, 367). One way to understand this effect is as a function of the relative exercise intensity-duration relationship (56, 134, 192, 367). Critical velocity (CV) or power (CP) occurs midway between the gas exchange threshold and $\text{VO}_{2\text{max}}$ and reflects the highest sustainable rate of aerobic metabolism (192). The maximal sustainable exercise time (t) at a given running velocity (V) can be estimated from: $t = D'/(V - \text{CV})$, where D' is the fixed distance that can be run above CV, fueled principally by anaerobic energy sources (192). Exercise intensity below CV can theoretically be sustained indefinitely, whereas exercise intensity above CV decreases in accordance with increasing anaerobic energy demands (134, 192). When the optimal racing strategy is to run (V) as close as possible to CV (121, 192, 193), any reduction in $\text{VO}_{2\text{max}}$ and increase in relative exercise intensity should reduce the lactate threshold and shorten the maximal sustainable exercise time (134). This may occur by a reduction in D' , CV, or some combination (192), though we are not aware of any experimental demonstration of this with dehydration. When examining self-paced exercise performance studies, it is important to remember that physiologic cues provide sensory feedback that limit performance (40, 198, 275). The higher relative effort ($\%\text{VO}_{2\text{max}}$) experienced when performing the same work while dehydrated is associated with higher ratings of perceived exertion (127, 224), thus performance may decline as voluntary effort is decreased (reduced V), a finding that has been repeatedly observed (Fig. 8).

It is also critical to remember several key methodological considerations. If circumstances do not require $V \geq \text{CV}$, then effort may be titrated more strategically and the effects of dehydration may not be observed. It must also be remembered that athletes will usually begin exercise well-hydrated. Dehydration takes time to develop (losses > intakes), thus exercise begun well-hydrated is unlikely to become impaired by dehydration unless the activity lasts 1 h or longer, assuming that sustainable exercise sweating rates are typically <1.5 L/h. Thus, a race just over 2-h duration may be compromised by dehydration only in the last portion of the event. If performance is tested or interpreted from changes in time for the entire race without this point in mind (144), the effects may seem small or nonexistent even though the direction of the effect is precisely in the direction of plausibility and most probably of great importance to competitors (177). Similarly, if the task is not long enough to produce dehydration $\geq 2\%$ of body mass, negative consequences should not be observed with greater than chance frequency (144). The method which employs dehydrating volunteers before testing, though

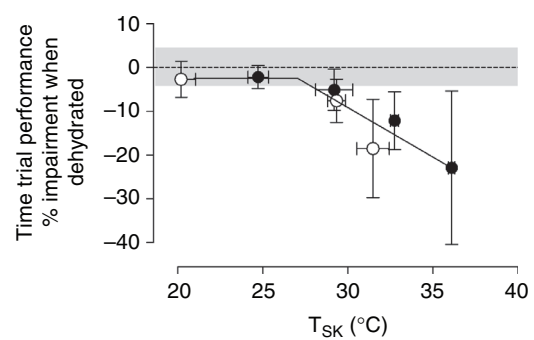


Figure 9 Endurance performance impairment when dehydrated by 3% to 4% body mass is a function of skin temperature (T_{sk}) above $\sim 27^{\circ}\text{C}$. Data are means [error bars are 95% CI] compiled from three studies (45, 52, 201) employing similar experimental procedures and time trial (TT) performance tests. Closed circles represent 15 min TT tests, open circles represent 30 min TT tests. All tests included a 30-min constant, submaximal exercise intensity preload. Shaded area represents $\pm 5\%$ coefficient of variation for time trial performance. Adapted, with permission, From Sawka et al. (310).

ecologically less valid, is probably the best means of studying this phenomenon scientifically since it is less confounded by a multitude of fatigue factors which the method minimizes.

Finally, the importance of the environment must be considered. The negative effects of dehydration on performance are generally larger in warmer environments (51, 52, 56, 201, 310). For example, Kenefick et al. (201) reported a 1.6% decline in the total work performed for every 1°C increase in skin temperature (T_{sk}) starting at a T_{sk} intercept of 29°C . At the same 4% level of dehydration, this amounted to a 3% impairment in aerobic time trial performance at an ambient temperature of 10°C , increasing to 23% at ambient temperature of 40°C . Nybo et al. (270) demonstrated that 4% dehydration reduced BV by $\sim 5\%$ ($\text{PV} \sim 10\%$) and lowered $\text{VO}_{2\text{max}}$ by 6% at T_{sk} 31°C and by 16% when T_{sk} was raised to 36°C using a water perfused suit. This strong modifying effect of T_{sk} on aerobic performance when dehydrated supports the proposed underlying skin blood flow demands and cardiovascular limitations on $\text{VO}_{2\text{max}}$, relative exercise intensity, and associated sensory cues (56). The quantification of the relationship between T_{sk} and aerobic performance impairment when dehydrated (4%) across a range of air temperatures ($2\text{--}40^{\circ}\text{C}$) is illustrated in Figure 9 (310).

Strength/power: Mechanisms of performance impairment

In 1923, Marriott (237) cited studies from the early 1900s when writing, “The withdrawal of fairly large amounts of water from the muscles (10–20% of their total water) fails to impair their function, as determined by electro-motor activity of the excised muscles.” The findings from Figure 8 largely corroborate that the effects of dehydration on strength or power are small (194) or nonexistent. Any definitive mechanism by which dehydration may impair strength or power *in vivo* is still unknown. Formulae provided by Pitcavage et al.

(284) allow an estimate for the reduction in intracellular muscle water associated with an increase in extracellular osmolality (tonicity). If Posm is taken simply as the product of plasma sodium $\times 2$, then a 10 mmol/kg increase in Posm should reduce muscle water marginally from 73% (well-hydrated) to 70% (dehydrated). This magnitude of change in Posm due to intracellular dehydration does not appear to alter metabolic energy stores (ATP/PC) (253), physiologic buffering capacity (32), muscle membrane excitability (69), thermal influences on muscle function/elastic energy (29), or any other aspect of intra-extracellular muscle compartment changes (103). The large osmotic stress that commonly accompanies high levels of intracellular dehydration (Fig. 8) can modulate opening of the blood-brain barrier (289), alter neuronal firing of hypothalamic osmoreceptor cells (33) and plausibly affect excitability of motor output pathways (101), but little has been revealed by studying the effects of dehydration on motor unit activation thus far (32, 103, 166, 195). Brain volume itself may shrink in response to severe and acute dehydration (155). However, the results of brain imaging studies indicate that mild dehydration ($\leq 3\%$ body mass) produced by fluid restriction and/or exercise may decrease (90), increase (199, 200, 346), or have no effect on brain volume (371). Brain volume is also protected by idiogenic osmole production when dehydration occurs gradually (155), thus changes in brain volume as a mechanism of subtle impairment in strength/power performance also seems unlikely.

Cognitive performance

Performance can also be viewed as an ability to carry out cognitive tasks. The potential effects of dehydration on cognitive performance are not extensively considered in this review, primarily because the plausibility that dehydration should alter cognition on physiologic grounds (osmolality and volume) appears small. Equivocal findings in this area could be due to common study-to-study differences, such as dehydration methods or the residual effects of dehydration which include fatigue or hyperthermia. In fact, reviews of the dehydration and cognition literature (146, 223) suggest that future research in this area should focus on the possible confounding effects of methods used to achieve dehydration (e.g., exercise heat exposure). Differences in the choice of cognitive test(s) or lack of test familiarization may also be blamed (30). Furthermore, investigations reporting significant alterations in cognition with dehydration as low as 1% to 2% of body mass (15, 61, 126, 142) did not establish a well-hydrated baseline.

Another important point not often considered is the absence of a genuine physiologic mechanism by which dehydration might impair cognitive function. In general, dehydration by $<2\%$ body mass would shrink the intravascular space by less than 200 ml and raise Posm by <5 mmol/kg. Much larger changes in Posm are required to increase blood-brain barrier permeability (289), which would not be possible with the levels of dehydration achieved in most of the dehydration

and cognition literature. Neural activity is closely related to cerebral blood flow (CBF) (133), but CBF is maintained at rest after intracellular (42) and extracellular (299) dehydration to 2% to 3% of body mass. Dehydration can reduce CBF transiently when an orthostatic challenge is imposed (42, 299) but this effect relates only to the risk for syncopal episodes in low tolerant individuals (290). As discussed above, the measured effects of mild dehydration on brain volume are also inconsistent.

Why might dehydration seem to impair cognition? It is generally acknowledged that dehydration has a negative effect on mood state and can additionally produce unpleasant symptoms such as dry mouth, thirst, and headache. Many studies investigating the impact of dehydration on cognition consistently report alterations in mood state such as perceived tiredness (348) and alertness (262) among other variables of mood state (e.g., fatigue, confusion, anger, and depression) (2, 15, 61, 100, 122, 126, 142, 276). The ability to overcome negative effects of stressors and maintain an effective level of performance is termed cognitive resiliency. This resiliency in cognitive performance has been reported even with dehydration to 4% of body mass and when exposed to cool, warm and hot environments (100). Negative mood states have also been reported to enlist greater effort (331) and it may be the case that individuals can elicit greater effort and detailed attention when in negative mood states such that decisions made were unaffected by dehydration. For example, Kempton et al. (2011) demonstrated that very mild dehydration ($<2\%$ body mass) did not impair cognitive performance or cerebral perfusion, but did increase measures of mental sedation. Dehydration also produced a stronger increase in the fronto-parietal blood oxygen-level-dependent response during an executive function task, suggesting that higher levels of neuronal activity were required (and called upon) to achieve the same performance. Thus, dehydration increased symptomologic distracters and required greater concentration to achieve the same level of performance. Symptomologic distracters are therefore a more probable explanation for impairment in cognitive function reported when dehydrated. Equivocal findings related to dehydration and cognition may therefore be more likely due to the variation in cognitive resilience among individuals.

Sport skill and task performance

Thus far, the impact of dehydration has been discussed relative to its effect on single energy systems and single performance outcome variables. Although laboratory exercise performance and “real life” (outdoor) sports performance bare an imperfect relationship, the impact of dehydration on performance in “real life” can be gauged through a limited body of sports performance literature. Many studies referenced on this topic report consistent negative effects of dehydration. This is true for a variety of sports, including tennis (178), soccer (1, 95, 240, 245, 305), basketball (19, 89), field hockey (231), cricket (83), wrestling (180, 203, 206, 373), boxing (335), rowing (333), and performance of a manual labor task (370).

In contrast, other studies report no impact of dehydration on performance of the same sports (43, 171, 233, 274, 332, 347).

The many difficulties in assessing genuine sport or task performance are no doubt a contributing factor to the inconsistency of findings in the literature. Given the methodological constraints of collecting data during actual competition, many of these investigations have assessed sport/task performance in laboratories or other controlled venues. In addition, other issues such as test familiarization, experience of the study volunteers, use of tests that are not ecologically valid, and other factors contribute to inconsistency in the literature and may limit the applicability of these studies to real-life performance. Lastly, it is also important to recognize that a very high degree of variation exists in performance of a sport skill or task (178) and that any alteration in performance could vary depending on the type of skill or task being performed.

Sport skill and task: Explanations of performance impairment

The majority of the studies on dehydration and sport skill or task performance assess one or some combination of the following: (1) aerobic endurance activity during simulated races, game play or performance of a work task; (2) motor skill function alone, or in combination with cognitive function, assessed by sport specific skill task or assessment of various measures of cognition or mood and; (3) anaerobic power or strength activity, measured by various sprinting and jumping tasks. Given the apparent complexity of many sport skill and other nonsport-related tasks, it is likely that they can be affected by an assortment of factors acting singly or in concert. Sport skill and related tasks are a complex make up of various components. These include aerobic endurance, specific expertise requiring both motor and cognitive function (e.g., ball dribbling, throwing, passing, shooting, and hitting) and anaerobic power or strength. Thus, the impact of dehydration will likely depend on the relative combination of factors that comprise the specific skill or task being performed, as well as the energy system predominantly tasked.

In one of the better examples of a field-valid study of endurance sport, Casa et al. (44) examined the impact of dehydration on trail running performance. Seventeen volunteers performed a simulated 12-km trail race while well-hydrated and dehydrated (~2% body mass loss). During the simulated 12-km trail race, volunteers continued to lose fluid via thermoregulatory sweating and completed the race with ~2% or ~4% dehydration, respectively. Performance time was ~5% slower when completing the race with 4% dehydration and volunteers reported a significantly greater perception of effort, thirst, and thermal sensation and greater feeling of tension, anxiety and fatigue. In another study of endurance, Wasterlund et al. (370) conducted one of the few studies directly measuring the impact of dehydration on a manual labor task during an actual work day. Performance was compared between those

who consumed fluid and those who did not. They reported that when subjects were dehydrated, productivity during an entire work day of stacking and debarking pulpwood was reduced by 12%.

Baker et al. conducted two well controlled studies investigating the impact of dehydration on basketball skill performance (19) and vigilance-related attention (18). A progressive decline in timed skills and shooting skills was reported as the level of dehydration increased (1-4%) in male basketball players. In addition, volunteers also reported feeling significantly more lightheaded and more leg, upper- and total body fatigue when 3% and 4% dehydrated (19). Baker et al. (18) also observed that when dehydrated (1-4%), vigilance-related attention was impaired by dehydration and volunteers felt more lightheaded, hot, overheated, and reported a greater feeling of overall body fatigue. Conversely, Hoffman et al. (171) observed the effect of ~2% dehydration on shooting skills, jump testing, and anaerobic power of young, male basketball players and reported no change in vertical jump height or shooting performance when dehydrated, but did observe a nonsignificant 19% decline in anaerobic power measured via a 30 s vertical jump test. In a study of upper body anaerobic power, Smith et al. (335) examined the impact of ~4% dehydration on boxing related performance (boxing ergometer) in eight young men. Smith et al. reported no changes in boxing performance when volunteers were dehydrated but also observed highly variable responses among individuals.

Overall it appears from observing this subset of studies (18, 19, 44, 171, 335, 370) on sport skill and task performance, and the literature in this area as a whole, that the impact of dehydration may be primarily dependent upon the make-up of the specific sport skill or task being studied. Tasks with a higher aerobic/endurance component such as trail running (44) or day-long manual labor (370) appear to be significantly impaired by dehydration which is supported by our analysis of the literature (Fig. 8). In regards to tasks that are predominantly strength or power reliant, the studies examining the impact of dehydration on lower body (171) and upper body (335) anaerobic power in basketball players and boxers both reported non-significant, but highly variable, effects of dehydration on performance. Sporting skill was (18, 19) or was not (171) impaired by dehydration.

Sport skill and task: Alternative explanations

It has been suggested that dehydration may negatively impact sport skill performance due to impaired balance, as body sway appears more pronounced after exercise and this effect is magnified when dehydration is present (82, 129). The possibility of impaired balance control associated with dehydration could be the result of changes in vestibular function or possibly vestibular afferent sensitivity (129, 220). However, the effect of dehydration on static balance is equivocal (82, 100, 129, 276). Derave et al. and Gauchard et al. (82, 129) demonstrated that exercise without fluid replacement decreased static balance ability; however, these findings were

likely confounded by local muscle fatigue brought about via previous exercise to elicit dehydration. In contrast, Patel et al. (276) demonstrated that ~2.5% dehydration did not affect static or dynamic balance. In addition, Ely et al. (100), using a 90-min recovery period following exercise-heat exposure to minimize the impact of recent exercise, reported that 4% dehydration did not alter measures of dynamic balance over a range of temperatures (10–40°C). However, when hyperthermia accompanies dehydration, dynamic balance becomes impaired (86).

In an effort to elucidate a possible physiologic mechanism behind impaired balance when dehydrated, Seay et al. (320) studied the impact of 3% extracellular dehydration (induced via diuretic) and 3–5% intracellular dehydration (induced via exercise in the heat) on standing balance. Seay et al. (320) hypothesized that either state might provide different neurogenic feedback to balance centers. Seay et al. reported that standing balance was not impaired by either intra- or extracellular dehydration and no relationships were apparent between physiologic changes due to PV loss or increases in Posm and balance performance. Taken together, these studies (100, 276, 320) would strongly suggest that any apparent changes in sport skill or task performance are not due exclusively to dehydration mediated alterations in standing or dynamic balance. Dehydration of 2% to 3% body mass has only been associated with transient changes in CBF (42, 299) and higher elevations in Posm are needed to alter the permeability of the blood-brain barrier (289) than those induced in most dehydration and sport skill or task performance studies. It seems likely that impaired balance when dehydrated is better explained by concurrent hyperthermia (86) or residual exercise fatigue (82, 129).

Abt et al. (1) hypothesized that “decreased mental performance” may lead to impaired peripheral motor control, thereby degrading skill performance. However there is little support that peripheral motor control is altered by dehydration. Numerous factors could serve to alter concentration and subsequently performance of a skill or task. Noakes and Edwards (96) proposed that performance may be impaired due to concurrent negative psychological associations attributed to thirst. However this assertion is not supported by earlier work by Edwards et al. (95) in which they observed the impact of dehydration on the performance of male soccer players during an intermittent running test used to simulate a competitive soccer match. Volunteers in this study performed the intermittent running test when well-hydrated, when 2% dehydrated and when dehydrated with repeated mouth rinsing to attenuate perceptions of thirst. Intermittent running performance was equally impaired in both the dehydrated and dehydrated + mouth rinse trials compared to control trials, suggesting that perception of thirst was not a factor.

There appears to be no genuine physiologic mechanism to explain alterations in sport skill, task or cognitive performance. However, as previously noted, factors associated with dehydration such as hyperthermia, glycogen depletion, hypoglycemia, mood state, discomfort, fatigue and greater

perceived exertion (heat, cold, headache, thirst, dry mouth, etc.) can serve as a distraction (e.g., reduced vigilance) (18) resulting in performance impairment of a particular sport skill or task. These negative alterations in mood state and attention have been reported in the cognitive literature as discussed in studies of basketball (18, 19), running (20, 44) and numerous other investigations. This may also be the case with the findings of many of the studies involving wrestling, where it is difficult to directly attribute negative findings of skill or task performance to dehydration alone given the stressful experience of prolonged exercise, food and fluid restrictions used to produce dehydration. It may also be the case that factors such as years of experience and competition, age, and skill level, may play an obvious role in the degree to which individuals may, or may not, be distracted by discomforts associated with dehydration.

Summary and Future Directions

Human performance is multifaceted and can be impacted by dehydration in numerous ways. Changes in volume and osmolality associated with dehydration clearly impact human physiology. A nonconventional analysis of the exercise performance literature revealed that the majority of studies support the concept that dehydration of $\geq 2\%$ loss in body mass negatively impacts endurance exercise performance while strength and power are negatively affected to a smaller degree. This is true whether individuals commence exercise in a dehydrated state, or accumulate fluid loss during the course of exercise. We propose that dehydration impairs endurance exercise performance primarily by reducing $\text{VO}_{2\text{max}}$ and increasing relative exercise intensity for any given task. This is directly the result of a reduced circulating BV. Mechanism(s) by which dehydration might impair strength or power are presently unclear. It appears that sports or tasks with a large endurance component will be more impacted by dehydration, while the performance of skills or tasks with a large anaerobic strength/power component appear to be more variable and influenced to a lesser degree. Skills or tasks with a large cognitive component, if impaired, are likely so as a result of changes in mood and the discomfort and distraction associated with dehydration. Future research directions might include studies designed to systematically test these conclusions.

Conclusion

The purpose of this paper was to present a unique and comprehensive review of the physiology, assessment, and performance effects of dehydration, particularly with reference to alterations in volume and osmolality. After thoroughly reviewing the dehydration literature, a number of important points mentioned in the paper stand out and stand repeating. Firstly, short-term, day-to-day body mass variation $\leq 1\%$ has long been recognized as synonymous with body water stability, as has the concept of establishing a well-hydrated baseline

prior to producing experimental dehydration. It is important to recognize that while the term “dehydration” represents a loss in TBW, two types of dehydration exist based on differences in accumulating solute losses. Intracellular dehydration is characterized by hyperosmolality of the ECF with modest ECF volume contraction. Extracellular dehydration is characterized by an unremarkable ECF osmolality but larger ECF volume contraction. When assessing dehydration, the adoption of biological variation and diagnostic accuracy (sensitivity and specificity) methods is essential. Our analysis of the literature supports the concept that $\geq 2\%$ body mass loss is the threshold for impairment of endurance exercise. One likely explanation for the impact of dehydration on endurance exercise is a reduction in $\text{VO}_{2\text{max}}$ associated with ECF volume contraction, which results in a higher relative effort ($\%\text{VO}_{2\text{max}}$) when performing the same work. At present there appears to be no clear mechanism by which strength and power exercise is altered by dehydration, but the degree of impairment is likely small. The potential for dehydration to impair cognition appears small and may be related simply to distraction or discomfort. The impact of dehydration on any particular sport skill or task is likely dependent upon the make-up of the task itself (aerobic, strength, cognitive, and motor skill).

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